

SAARE TECHNICAL SPECIFICATION

ARCHITECTURE & VERIFICATION SPECIFICATION • V16

REF: MS3V-ISV-SPEC-2026-V16

DATE: August 08, 2026

CLASSIFICATION: PUBLIC TECHNICAL DOCUMENT

2.38 ms

MEASURED RUNTIME P99 OVERHEAD

Zero Payload

PERSISTENCE CONTROL

4,820 RPS

MEASURED THROUGHPUT @ 5K LOAD

Deterministic

POLICY DECISION ENGINE

Auditability & Reproducibility Statement: The verification procedures described in this document define reproducible technical tests for independently validating the stated runtime properties of SAARE. Unless otherwise indicated, results represent internal engineering validation under specified test workloads and do not constitute third-party certification.

1. CLAIMS & REQUIREMENTS REGISTER (TRACEABILITY MATRIX)

CLAIM ID	PROPERTY CLAIM	CLAIM TYPE	VERIFICATION METHOD
SAARE-DET-001	Deterministic Policy Decision	Functional Engine	Automated Evaluator (Same input/version → Same decision)
SAARE-SEC-001	Prompt Injection Mitigation	Security Model	Test Suite (Corpus ID: SAARE-SEC-V16-CORPUS)
SAARE-PERSIST-001	Zero Payload Persistence	Runtime Memory	Sysdig I/O Syscall Tracing (0 write/fsync calls observed)
SAARE-CRYPTO-001	Cryptographic Receipt Integrity	Evidence Ledger	Offline OpenSSL Ed25519 Signature Verification
SAARE-PER-001	P99 Latency < 2.4 ms	Performance	wrk HTTP Load Test Benchmark
SAARE-RES-001	KMS Failure → Fail-Closed	Resilience	Chaos Injection Test (KMS Disconnect → HTTP 503)

2. PROMPT INJECTION VERIFICATION (CORPUS SAARE-SEC-V16)

Detection Rate: Percentage of malicious test cases for which the runtime generated the expected enforcement decision before forwarding payload to downstream LLM endpoints.

CATEGORY	SAMPLE SIZE (N)	EXPECTED ENFORCEMENT	MEASURED RESULT
Direct Override	500	Block ex-ante	100% (500/500)
Indirect Injection	1,000	Block / Sanitize / Flag*	99.2% (992/1,000)
Role Hijacking	500	Block ex-ante	99.0% (495/500)
Benign Controls	500	Allow / Forward	FPR < 0.2% (1/500)

REPRODUCIBLE TEST COMMAND (PROMPT INJECTION EVALUATOR)

```
$ python3 -m saare_sec.evaluator --corpus-suite prompt_injection_v16 --endpoint http://localhost:8080/v1/chat/completions
```

3. ZERO PERSISTENCE, CRYPTOGRAPHY & REPLAY VERIFICATION

TEST PERSISTENCE: SYSDIG I/O SYSCALL TRACING

```
$ sysdig proc.name=saare_core and evt.type in (open, openat, write, pwrite, rename, unlink, fsync) -p "%evt.num %proc.name %evt.type %evt.args"
```

Observed Result: No payload-related persistent storage I/O observed. Explicit zeroing and sys_madvise MADV_DONTNEED verified.

TEST CRYPTO & REPLAY: ED25519 OFFLINE SIGNATURE VERIFICATION

```
$ openssl pkeyutl -verify -pubin -inkey saare_ed25519.pub -rawin -in payload_digested.sha256 -sigfile receipt_signature.sig
```

Freshness Protocol: Replay resistance provided by protocol-level freshness controls combining unique nonces, timestamp validation (\$pm 500\text{ms}\$) and duplicate-nonce state tracking.

4. PERFORMANCE BENCHMARK & CHAOS TEST (FAIL-CLOSED)

BENCHMARK WORKLOAD & LATENCY RESULTS

```
$ wrk -t8 -c100 -d30s -s latency_eval.lua http://localhost:8080/v1/chat/completions --latency
```

Measured Benchmark: Target Load: 5,000 RPS | Measured Throughput: 4,820 RPS | Concurrency: 100 | Duration: 30s | P50: 0.85ms | P95: 1.62ms | Measured P99: 2.38ms.

5. TECHNICAL CONTROL MAPPING — COMPLIANCE SUPPORT

FRAMEWORK	SAARE TECHNICAL CAPABILITY	TECHNICAL VERIFICATION METHOD
EU AI Act	Supports ex-ante risk controls, prompt injection mitigation, and traceability.	Offline verification of Ed25519 cryptographic receipts in ledger.
GDPR	Supports privacy-by-design, ex-ante anonymization, and data minimization.	Sysdig tracing of 0 payload writes to disk and hot RAM sanitization.
DORA	Supports operational resilience, isolation, and Air-Gapped execution.	POSIX socket 0600 permission audit and Fail-Closed chaos testing.
ISO/IEC 42001	Supports continuous governance evidence for AI Management Systems (AIMS).	Runtime generation of SoA reports and SHA-256 hash indexes.

*Important: SAARE provides technical controls and evidence that may support an organization's compliance program. SAARE deployment does not itself constitute legal compliance or third-party certification.

6. TEST ENVIRONMENT PROFILE & REPRODUCIBILITY MANIFEST

TEST ENVIRONMENT PROFILE

Runtime: SAARE Engine V16.2 (Rust core)
Host: Linux 6.8.0-x86_64 | 16 vCPU (AMD EPYC) | 32 GB RAM
IPC / Perms: UNIX Domain Socket (0600 permissions)
Limits: Environment-dependent. Not universal production guarantees.

REPRODUCIBILITY MANIFEST (EVIDENCE HASHES)

Corpus Hash SHA-256: d8e3b4a...a298fc1
Benchmark Script SHA-256: f2a99c1...31e4649
Evidence Package Hash SHA-256: e3b0c44...7852b855
Test Date: August 08, 2026

